YASH SRIVASTAVA

Canberra, ACT | yashsrivastava1998@gmail.com | +61-0460339056 | linkedin.com/in/yash9988/ | github.com/Yash9988

EDUCATION

Master of Computing in 'Data Science and Machine Learning' *Canberra, Australia*
Australian National University - **5.94/7** CGPA July, 2022 – June, 2024

Bachelor of Technology in 'Computer Science & Engineering' *Chennai, India*
SRM Institute of Science and Technology - **8.48/10** CGPA July, 2016 – June, 2020

SKILLS

- **Tools**: Git, Power BI, Kafka, Jira, PyCharm, Jupyter, Docker, etc.
- **Languages**: Python, R, Java, C++, SQL, HTML, CSS, JavaScript, Shell Scripting.
- **Frameworks/Libraries/Databases:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, OpenCV, TensorFlow, PyTorch, PySpark, MySQL, PostgreSQL, MongoDB, Linux.
- **Concepts:** Machine Learning, Artificial Intelligence, Reinforcement Learning, DNN, NLP, LLM, Computer Vision, Time Series Analysis, Real-time Data Analysis, Data Visualization, Statistical Modelling, Model Evaluation.

WORK EXPERIENCE

Australian National University *Canberra, Australia*
Teaching Associate July, 2023 – Current

- The role involved synthesizing data and information into effective assessments and study materials for graduate courses in Machine Learning and Structured Programming.
- Developed comprehensive course content through practical evaluation of assignments and feedback from students. Leveraged team efforts on course projects, ensuring students achieved tangible learning outcomes.
- Helped improve student performance through targeted assessment strategies and mentorship, contributing to the development of effective teaching materials for courses at ANU.

Australian Plant Phenomics Network *Canberra, Australia*
Computer Vision Engineer February, 2024 – June, 2024

- The task involved using advanced imaging technologies to accurately identify and analyze food-grade recombinant protein expression in plant leaves.
- Utilized 3D hyperspectral scanners, thermal cameras, and RGBD cameras for precise data capture. Developed machine learning models to automatically identify and segment leaves, enhancing the accuracy of protein expression analysis through calibrated imaging equipment and collected RGB images.
- The project significantly improved the efficiency and accuracy of identifying food-grade recombinant proteins in plant leaves. Advanced imaging techniques and machine learning models contributed to more reliable agricultural research, leading to enhanced understanding and control over protein expression.

Australian National University *Canberra, Australia*
Team Lead & Researcher July, 2023 – November, 2023

- The goal was to develop a PIR sensor-based system for 3D-space mapping and movement tracking.
- Leveraged real-time data analysis techniques to enhance user experience through the PIR sensor-based system. Developed algorithms for 3D-space mapping and implemented machine learning models to improve spatial awareness and decision-making processes, such as flow planning.
- The project resulted in reduced congestion, efficient handling, empowered decision-makers, and enhanced safety in spatial planning and security systems. Real-time data analysis and enhanced user experience contributed significantly to the success of the PIR sensor-based system.

AlgoT *Gurugram, India*
Data Scientist August, 2021 – June, 2022

- Developed an automated stock trading algorithm using ensemble machine learning models to predict future market trends.
- Utilized a combination of statistical analysis under mentorship from a veteran data scientist. Deployed models with Python, Scikit-learn, and TensorFlow frameworks to automate trading processes efficiently.
- Achieved significant improvements in trading efficiency by deploying robust algorithms that effectively predicted market trends. The project contributed valuable insights into automated trading strategies.

ADDITIONAL EXPERIENCE

Alphabt-TVS Motors *Hosur, India*
Computer Vision Intern June, 2019 – July, 2019

- Resolved the problem of serial code mismatches on chassis parts and vehicles through computer vision techniques.
- Integrated IoT applications with deep learning frameworks like CNNs using Python libraries such as Keras, OpenCV, NumPy, CNTK, and TensorFlow to detect and localize engraved digits accurately. Cross-analyzed the model's predictions against company databases to ensure accurate part registration.
- Reduced serial code mismatches significantly by automating the detection process. The project improved manufacturing efficiency and accuracy in vehicle assembly processes.

Lucid Software Limited *Chennai, India*
Computer Vision Intern June, 2018 – July, 2018

- Implementation of automated defect detection using computer vision techniques in a research setting.
- Developed convolutional neural networks (CNNs) using Python with Keras and other libraries to analyze x-ray images for defect localization on welded surfaces. Experimented with various CNN architectures, optimizing them for accuracy and speed.
- Improved the precision of defect detection in welds by creating a reliable automated system that reduced manual inspection requirements. The project contributed to enhancing product quality control processes at the company.

The Boring Company (Team Badgermole)
Software Developer June, 2020 – September, 2020

- Design an efficient monitoring & communication software for a semi-autonomous Tunnel Boring machine.
- Developed a GUI for controlling and monitoring the machine and designed communication channels to ensure constant contact among all devices. Utilized engineering principles and cutting-edge technology to design and control the tunnel boring machine.
- Qualified for the final round, competing alongside 11 other prestigious teams worldwide. The project demonstrated innovative engineering skills and contributed to the advancement of tunnel boring technologies, showcasing efficient semi-autonomous solutions.

NextTech Labs *Chennai, India*
Artificial Intelligence Researcher April, 2017 – July, 2018

- Worked on ML research, applying Deep Learning frameworks to solve computer vision and natural language processing problems.
- Utilized deep learning frameworks such as TensorFlow and Keras to develop models for solving complex problems in computer vision and natural language processing. Participated in hackathons and seminars to stay updated with the latest trends and technologies in AI research.
- Enhanced problem-solving abilities through participation in hackathons and seminars. Contributed to innovative solutions in machine learning, demonstrating a strong foundation in deep learning techniques for both computer vision and natural language processing tasks.

PROJECTS

- **Time Series Forecasting using ARIMA:** Developed a model to predict future trends using historical data (e.g., stock prices).
- **Breast Cancer Detection using ML:** Built a machine learning model to classify breast cancer images as benign or malignant
- **Chat-Bot:** Developed a conversational chatbot capable of understanding and responding to user queries.
- **Hand & Pose Estimation:** Implemented a system to detect and track human hands and estimate their pose in real-time.
- **Reinforced Learning - Flappy Bird Game:** Created an AI agent that learned to play Flappy Bird through trial and error, demonstrating reinforcement learning principles.

PUBLICATIONS

Enhanced Image Vision and Resolution during Low Light Conditions using GANs - International Journal of Recent Technology and Engineering (IJRTE), May 2020, Co-author.